

MATH 3310 HOMEWORK ASSIGNMENT 2

DUE ON THURSDAY 8 SEPTEMBER 2022

Please turn in your homework in class or to Weeks Hall 347 no later than 5 pm.

- (1) Consider the sets

$$U = \{x \in \mathbb{R} \mid |x - 2| \leq 5\},$$

$$V = \{x \in \mathbb{R} \mid |x + 1| < 3\}, \quad \text{and}$$

$$W = \{x \in \mathbb{R} \mid |x| > 2\}.$$

Use the interval notation to express the following sets:

(a) U .

(d) $U \cap V$.

(b) V .

(e) $V \cup W$.

(c) W .

(f) $U - W$.

- (2) For every real number x set

$$A_x = \{x^2\}, \quad B_x = (-\infty, x), \quad \text{and} \quad C_x = [x - 1, x + 1].$$

For $X = \{0, 1, \pi, 5\}$ determine each of the following sets.

(a) $\bigcap_{x \in X} A_x$ and $\bigcup_{x \in X} A_x$.

(b) $\bigcap_{x \in X} B_x$ and $\bigcup_{x \in X} B_x$.

(c) $\bigcap_{x \in X} C_x$ and $\bigcup_{x \in X} C_x$.

- (3) Determine the set A_n in each of the following collections:

(a) $\{A_n\}_{n \in \mathbb{N}} = \{[0, 1], [0, 2], [0, 3], \dots\}$.

(b) $\{A_n\}_{n \in \mathbb{N}} = \{[1, 3], [1, 2 + \frac{1}{2}], [1, 2 + \frac{1}{3}], \dots\}$.

(c) $\{A_n\}_{n \in \mathbb{N}} = \{(-\frac{2}{3}, 3), (-\frac{5}{3}, 6), (-\frac{8}{3}, 9), \dots\}$.

- (4) Consider the set $A = \{a, b, c, d, e, f, g\}$. For each of the following collections of subsets of A decide if it is a partition of A , and if it is not explain why?

(a) $\{\{ \}, \{a, e, f\}, \{b, c, d, g\}\}$.

(d) $\{\{a, g, b\}, \{e, f\}, \{b, c, d\}\}$.

(b) $\{\{a, e\}, \{b, d\}, \{f, g, c\}\}$.

(e) $\{\{a, e\}, \{b, c, d, f, g\}\}$.

(c) $\{A, \{a, b, c, d, e, f, g\}\}$.

(f) $\{\{a, b\}, \{c, d\}, \{f, g\}\}$.

- (5) Consider the set $A = \{a, b, c, d, e, f, g, h, i, k, l\}$ and give an example of a partition $\mathcal{S}$ of A that satisfies

- $|\mathcal{S}| = 5$.

- $|X| \neq 3$ for all $X \in \mathcal{S}$.

- There is a subset $\mathcal{T}$ of $\mathcal{S}$ with $|\mathcal{T}| = 4$ and $|\bigcup_{X \in \mathcal{T}} X| = 9$.